SEMESTER 2 • ENGINEERING SCIENCE • ELECTRICAL + ELECTRONICS HANDBOOK

# Fundamentals of Electrical and Electronics Engineering

A complete student textbook for DC circuits, AC systems, residential wiring, AC power, electricity bill, safety, passive components, diodes, rectifiers, transistors and logic gates with formulas, circuit diagrams, solved numericals, truth tables and full answers.

COURSE CODE
**2031**

COURSE TITLE
**Fundamentals of Electrical and Electronics Engineering**

PROGRAMME
**Computer / Electrical / Electronics / allied diploma programs**

SEMESTER
**2**

CREDITS
**3**

CATEGORY
**Engineering Science**

PERIODS / WEEK
**3 (L:2, T:1, P:0)**

PERIODS / SEMESTER
**45**

**45**

## Periods of core fundamentals

This course builds the foundation needed for circuits, electronics, wiring, power systems, instruments, automation and digital logic.

COURSE OVERVIEW

# What this subject teaches

Fundamentals of Electrical and Electronics Engineering gives a common foundation for many diploma branches. It explains how voltage, current and resistance behave in circuits, how AC voltage is generated, how residential wiring is protected, how power and energy bills are calculated, and how passive and active electronic components work.

## Electrical foundation

Students learn current, voltage, resistance, power, Ohm’s law, resistance combinations, AC waveform terms and power calculations.

> Electrical circuit □□□□□□□□□□□□□□ voltage, current, resistance □□□□□□□□□□□ □□□□□□ □□□□□□□.

## Residential and safety awareness

Service connection, fuses, energy meter, main switch, MCB, ELCB, neutral link, earthing and basic safety.

## Electronics foundation

Identify resistors, capacitors, inductors, transformers, diodes, transistors and logic gates and understand applications.

HOW TO USE THIS HANDBOOK

# Study method

### 1. Read circuit

Identify source, load, switch, resistor, diode or transistor.

### 2. Write formulas

Use symbols and units correctly.

### 3. Solve steps

Given data, required value, formula, substitution, answer and unit.

### 4. Practise diagrams

Draw symbols, rectifier waveforms and truth tables neatly.

**Electrical Safety:** Never practise wiring on live supply. Use correct protective devices and proper earthing.

TABLE OF CONTENTS

## Clickable navigation


- Objectives & Outcomes →
- Roadmap →
- Module 1 →
- Module 2 →
- Module 3 →
- Module 4 →
- Formula Bank →
- Units Bank →
- Circuit Bank →
- Component Bank →
- Solved Numericals →
- Expected Questions →
- Practice →
- Quick Revision →
- Model Paper →
- Solutions →


COURSE DETAILS

# Objectives, prerequisites, outcomes and CO-PO mapping

## Course Objectives

1. To provide basic knowledge of the different elements and concepts of electrical engineering.
2. To imbibe basic concepts of various active and passive electronic components.

## Prerequisites

- Basic knowledge in Physics — Applied Physics I, Semester 1.
- Basic knowledge in Mathematics — Engineering Mathematics I, Semester 1.

> Ohm’s law, power, units, algebra, ratio □□□□□□ □□□□□□□□□□□□□□□□□□ □ subject □□□□□□□□□□□.

## Course Outcomes

| CO | Description | Duration | Level | Student meaning |
|---|---|---|---|---|
| CO1 | Identify various combinations of resistors and basic terms in AC systems. | 12 h | Applying | Solve resistor circuits and AC waveform terms. |
| CO2 | Solve various powers in AC circuits and calculate monthly electricity bill. | 10 h | Applying | Use power factor, P/Q/S and kWh billing. |
| CO3 | Identify passive components, colour coding and applications. | 10 h | Applying | Identify resistors, capacitors, inductors and transformers. |
| CO4 | Summarize diodes, transistors and logic gates. | 11 h | Understanding | Understand rectifiers, zener diode, transistor amplifier and logic gates. |
| Series Test | Series Test | 2 h | - | Series Test I after Module 2 and Series Test II after Module 4. |

## CO1 → PO1

3, strongly mapped.

### CO2 → PO1

3, strongly mapped.

### CO3 → PO1

3, strongly mapped.

### CO4 → PO1

2, moderately mapped.

**Scale:** 3 - Strongly mapped, 2 - Moderately mapped, 1 - Weakly mapped.

ROADMAP

## 45-period learning plan

| Module | Duration | Main topics | Calculation focus | Diagram focus |
|---|---|---|---|---|
| 1. DC Circuits and Fundamentals of AC Systems | 12 h | Ohm's law, resistance combinations, EMI, AC terms | V=IR, P, Req, f,T,ω | DC, series/parallel, AC waveform |
| 2. Residential Wiring, AC Power, Energy Bill and Safety | 10 h + Test I | Wiring, protection, AC power, energy bill | Power factor, P/Q/S, kWh | Wiring block, power triangle |
| 3. Passive Electronic Components | 10 h | Resistors, capacitors, inductors, transformers | Colour code, capacitance, turns ratio | Symbols, charging, transformer |
| 4. Diodes, Transistors and Logic Gates | 11 h + Test II | Diodes, rectifiers, zener, transistor, gates | Truth tables, waveform reading | Rectifiers, transistor, logic gates |

MODULE 1 • CO1

# DC Circuits and Fundamentals of AC Systems

Current, voltage, resistance, power, Ohm’s law, laws of resistance, DC/AC circuits, resistor combinations, electromagnetic induction, AC voltage generation and AC terms.

## Module Outcomes

- Apply Ohm’s law.
- Solve resistor combinations.
- Summarize electromagnetic induction and AC generation.
- Illustrate AC terms and solve problems.

## Electricity basics

Electricity is a form of energy associated with electric charge. Current is charge flow, voltage drives current, resistance opposes current and power is rate of electrical work.

> Current flow □□□□□□□□□ voltage □□□□. Resistance current-□□ oppose □□□□□□□□.

| Quantity | Meaning | Symbol | Unit |
|---|---|---|---|
| Current | Rate of charge flow | I | A |
| Voltage | Potential difference | V | V |
| Resistance | Opposition to current | R | Ω |
| Power | Rate of doing electrical work | P | W |

## Ohm’s law and laws of resistance

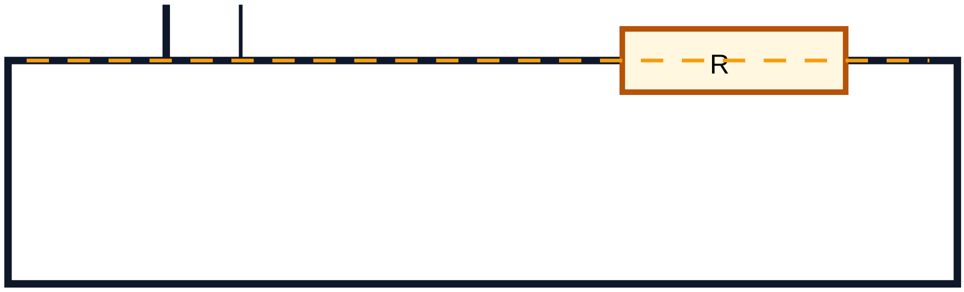
At constant temperature, current through a conductor is directly proportional to voltage across it.

$$V = IR,\ I = V/R,\ R = V/I$$

$$P = VI = I^2R = V^2/R$$

$$R = \rho l/A$$

## Simple DC circuit




**Closed DC circuit with load resistor**

## Series and parallel

Series: same current, voltage divides. Parallel: same voltage, current divides.

Series: Req = R1 + R2 + R3 + ...

Parallel: 1/Req = 1/R1 + 1/R2 + ...

## Equivalent resistance derivation

### Series

V=V1+V2+V3; IR=IR1+IR2+IR3; therefore Req=R1+R2+R3.

### Parallel

I=I1+I2+I3; V/Req=V/R1+V/R2+V/R3; therefore reciprocal relation.

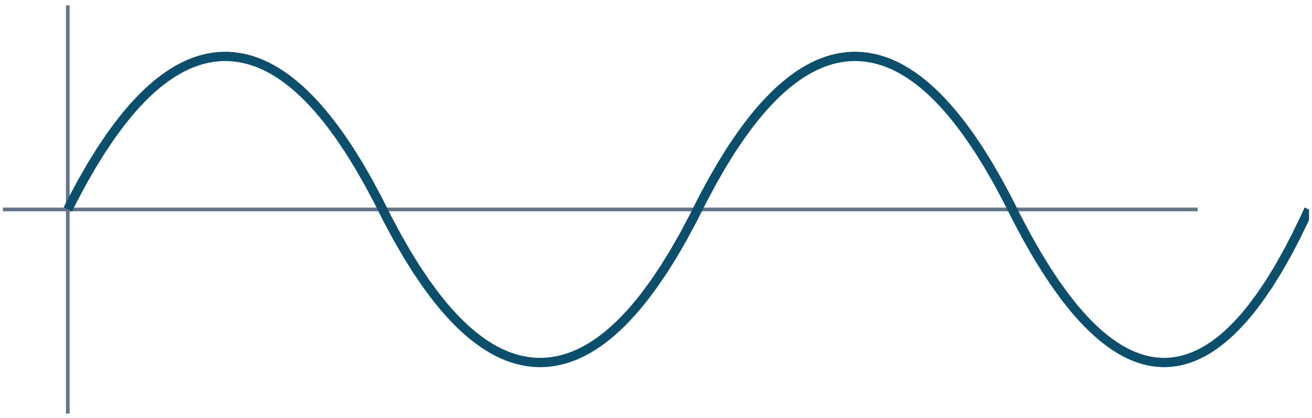
## AC systems

AC changes magnitude and direction periodically. AC voltage is generated when a coil rotates in a magnetic field and magnetic flux linked with it changes. This is based on electromagnetic induction.

> Generator-□ coil magnetic field-□ rotate □□□□□□□□□□□ voltage induce □□□□□□□. □□□□□□□□□ AC

waveform □□□□□□□□.

$$v = V_m \sin(\omega t),\ \omega = 2\pi f,\ T = 1/f,\ f = 1/T$$



**AC voltage waveform**

## Important AC terms

- Cycle: one complete positive and negative variation.
- Frequency: cycles per second in hertz.
- Time period: time for one cycle.
- Amplitude/maximum value: peak value.
- Instantaneous value: value at a particular instant.
- Reactance: opposition to AC due to L or C.
- Impedance: total opposition to AC.

**Solved examples**

1) $R=10\ \Omega$, $I=2$ A. $V=IR=2\times10=20$ V.
2) 6 Ω and 3 Ω in series: $R_{eq}=9\ \Omega$. In parallel: $1/R=1/6+1/3=1/2$, $R=2\ \Omega$.
3) $f=50$ Hz. $T=1/f=0.02$ s; $\omega=2\pi f=314$ rad/s.

MODULE 2 • CO2

# Residential Wiring, AC Power, Energy Bill and Safety

## Module Outcomes

- Outline residential wiring requirements.
- Identify powers in AC circuits.
- Calculate monthly electricity bill.
- Summarize safety precautions.

## Residential wiring

Residential wiring distributes electrical energy safely from service connection to loads like lights, fans, sockets and appliances.

> House wiring-□ supply load-□□□□□□□ □□□□□□□□□□□□□□□□□ protection devices □□□□ □□□□.

## Open conduit wiring

Conduit is visible on wall surface; easier maintenance.

## Concealed conduit wiring

Conduit is hidden inside wall; better appearance, careful planning needed.

## Wiring materials and devices

### Service connection

Connection from supply authority to consumer.

## Cut-out fuse

Melts during excessive current.

## Energy meter

Measures kWh energy consumed.

## Main switch

Controls entire installation.

## MCB

Protects overload and short circuit.

## ELCB

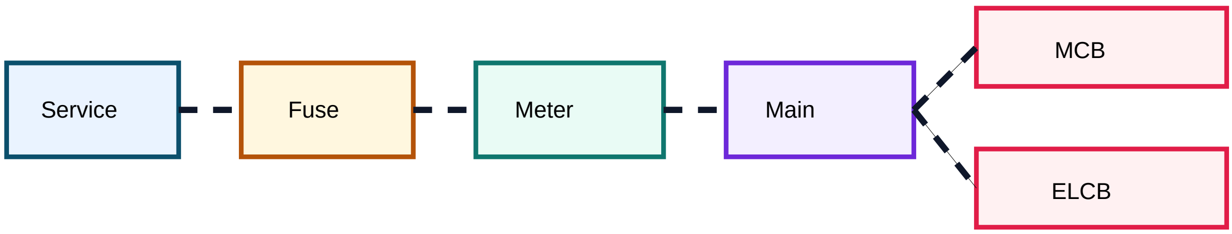
Protects against earth leakage shock hazard.

## Neutral link

Common neutral connection point.

## Earthing

Low resistance fault path to ground.




**Basic residential wiring protection chain**

## AC power

### Power factor

Cosine of phase angle between voltage and current.

$$\text{Power factor} = \cos\varphi$$

### Active power

$$P = VI\cos\varphi \text{ (W)}$$

### Reactive power

$$Q = VI\sin\varphi \text{ (VAR)}$$

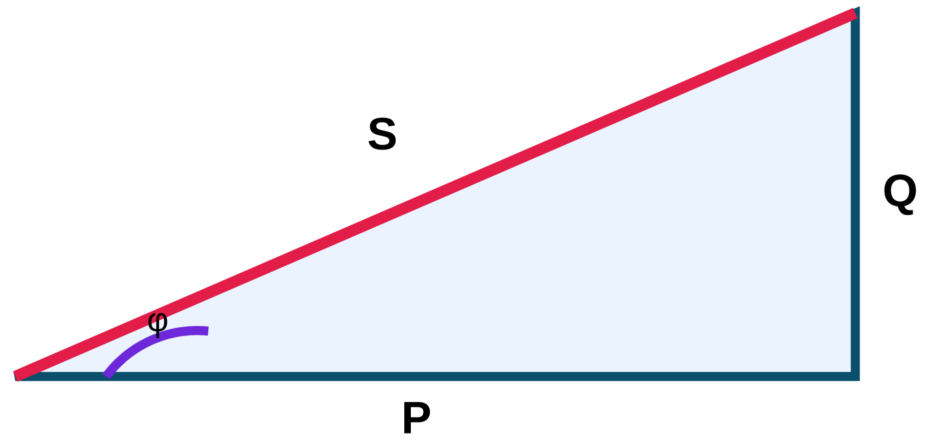
### Apparent power

$$S = VI \text{ (VA)}$$

Single phase power: $P = VI\cos\varphi$

Three phase power: $P = \sqrt{3}\, V_L I_L \cos\varphi$

Energy = Power × Time; 1 unit = 1 kWh; Bill = units × rate + fixed charges




**Solved examples**

1) V=230 V, I=5 A, pf=0.8. S=VI=1150 VA; P=VIcosφ=920 W; if sinφ=0.6, Q=690 VAR.
2) 1000 W heater used 2 h daily for 30 days at ₹6/unit plus ₹50 fixed. Energy=1 kW×60 h=60 units; charge=60×6+50=₹410.
3) Three phase: VL=400 V, IL=10 A, pf=0.8. P=√3×400×10×0.8=5.54 kW.

**Safety:** Switch off supply, use proper fuse/MCB/ELCB, avoid wet hands, provide earthing, do not overload sockets, replace damaged insulation.

MODULE 3 • CO3

# Passive Electronic Components

## Module Outcomes

- Identify resistors and colour coding.
- Identify capacitors and combinations.
- Explain inductance and inductor applications.
- Explain transformer working.

## Active vs passive

Passive components do not provide power gain. Examples: resistor, capacitor, inductor and transformer. Active components can control current or amplify signals.

> Passive component power gain □□□□□□□. Active component amplification/switching □□□□□□□.

## Resistors

Resistors limit current, divide voltage, set bias, provide pull-up/pull-down and form timing networks.

### Fixed

Fixed value, colour-coded or printed.

### Variable

Adjustable resistance for control/calibration.

### Carbon / wire wound

Carbon for general use; wire wound for higher power.

## Colour code

| Colour | Digit | Multiplier |
| --- | --- | --- |
| Black | 0 | $10^0$ |
| Brown | 1 | $10^1$ |
| Red | 2 | $10^2$ |
| Orange | 3 | $10^3$ |
| Yellow | 4 | $10^4$ |
| Green | 5 | $10^5$ |
| Blue | 6 | $10^6$ |
| Violet | 7 | $10^7$ |
| Grey | 8 | $10^8$ |
| White | 9 | $10^9$ |
| Gold | - | ±5% tolerance |
| Silver | - | ±10% tolerance |

**Colour code example**

Brown, Black, Red, Gold: digits 1,0; multiplier $10^2$; value = $10 \times 10^2 = 1000\ \Omega = 1\ k\Omega$; tolerance ±5%.

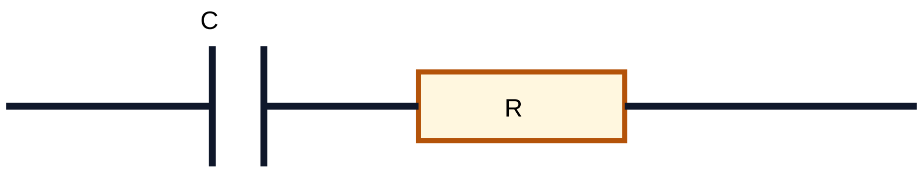
## Capacitors

A capacitor stores charge and energy in an electric field. Used for filtering, coupling, decoupling, timing, tuning and smoothing rectifier output.

> Capacitor charge store □□□□□□□□. Supply remove □□□□□□□□ load □□□ discharge □□□□□□□□.

$$\text{Parallel capacitors: } C_{eq} = C_1 + C_2 + C_3 + \ldots$$

$$\text{Series capacitors: } 1/C_{eq} = 1/C_1 + 1/C_2 + \ldots$$




**Capacitor charging/discharging circuit**

**Capacitance example**

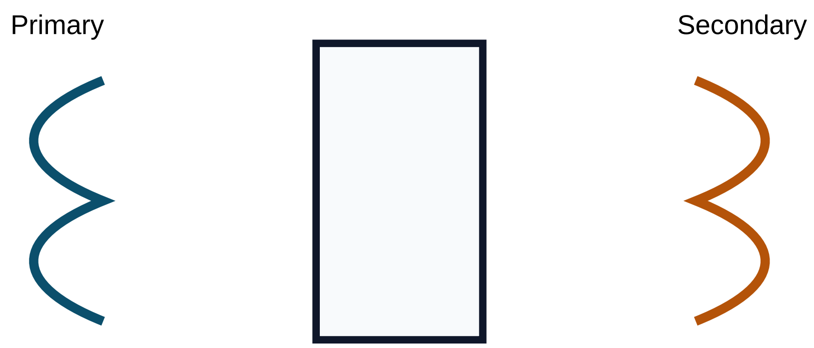
6 μF and 3 μF: parallel C=9 μF. Series: 1/C=1/6+1/3=1/2, so C=2 μF.

## Inductors and transformers

Inductor stores energy in magnetic field. Self inductance opposes change in its own current. Mutual inductance induces emf in nearby coil. Transformer works on mutual induction.

$$\text{Turns ratio} = N2/N1 = V2/V1$$

$$\text{Ideal transformer: } Vp/Vs = Np/Ns$$




**Transformer: mutual induction**

MODULE 4 • CO4

# Diodes, Transistors and Logic Gates

## Module Outcomes

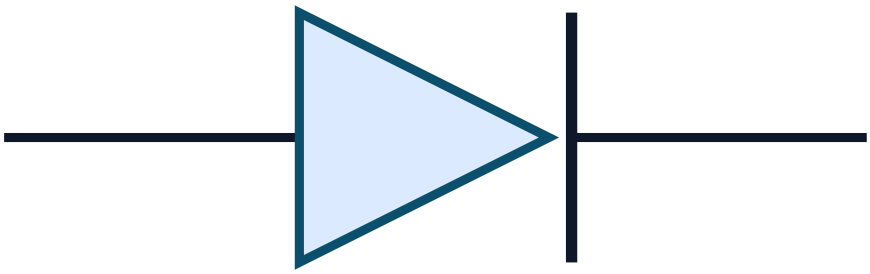
- Illustrate rectifier circuits.
- Explain zener diode.
- Summarize transistor as amplifier.
- Outline logic gates and applications.

## PN junction diode

A diode conducts mainly in forward bias and blocks in reverse bias except leakage current.

> Diode one-way valve □□□□. Forward bias current □□□□□, reverse bias □□□□□.

## Diode and rectifiers



**PN junction diode symbol**

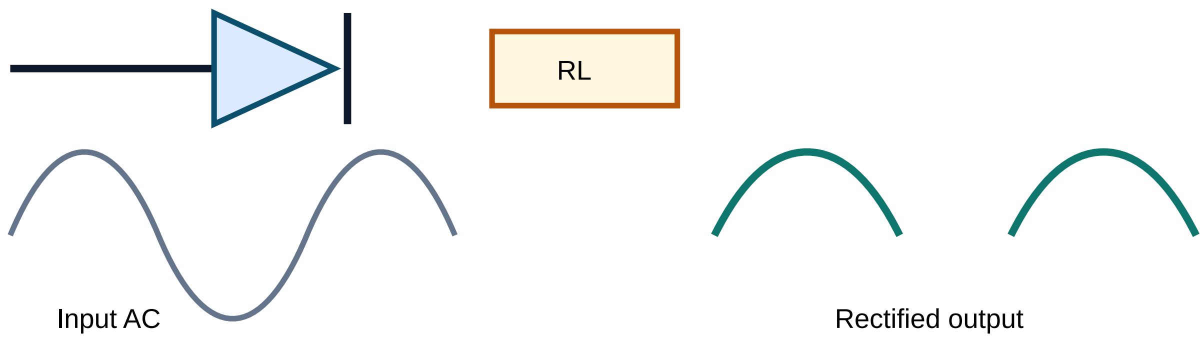
Rectifier converts AC into pulsating DC. Half wave uses one diode. Full wave centre tapped uses two diodes and centre-tapped transformer. Bridge rectifier uses four diodes and no centre tap.

## Half wave rectifier

Only one half cycle appears across the load.

### Bridge rectifier

Both half cycles are used; output ripple frequency is higher than half wave.




## Zener diode and transistor

### Zener diode

Works in reverse breakdown region and maintains nearly constant voltage. Used as voltage regulator and protection device.

### Transistor amplifier

A small base input controls larger collector current, producing amplified output.

> Base-□ □□□□□□ signal □□□□□□□□□□ collector current □□□□ □□□□□□□□ □□□□□□.

## Logic gates and truth tables

### AND

| A | B | Y |
|---|---|---|
| 0 | 0 | 0 |
| 0 | 1 | 0 |
| 1 | 0 | 0 |

| | | |
|---|---|---|
| 1 | 1 | 1 |

## OR

| A | B | Y |
|---|---|---|
| 0 | 0 | 0 |
| 0 | 1 | 1 |
| 1 | 0 | 1 |
| 1 | 1 | 1 |

## NOT

| A | Y |
|---|---|
| 0 | 1 |
| 1 | 0 |

## NAND

| A | B | Y |
|---|---|---|
| 0 | 0 | 1 |
| 0 | 1 | 1 |
| 1 | 0 | 1 |
| 1 | 1 | 0 |

## NOR

| A | B | Y |
|---|---|---|
| 0 | 0 | 1 |

| | | |
|---|---|---|
| 0 | 1 | 0 |
| 1 | 0 | 0 |
| 1 | 1 | 0 |

### XOR

| A | B | Y |
|---|---|---|
| 0 | 0 | 0 |
| 0 | 1 | 1 |
| 1 | 0 | 1 |
| 1 | 1 | 0 |

**NAND and NOR are universal gates.** Logic gates are used in computers, calculators, alarms, control circuits and automation.

FORMULA BANK

## Important formulas

### Basic Electrical

$V=IR$, $I=V/R$, $R=V/I$

$P=VI=I^2R=V^2/R$

$R=\rho l/A$

### Resistance and AC

Series: $R_{eq}=R1+R2+R3+...$

Parallel: $1/R_{eq}=1/R_1+1/R_2+...$

$v=V_m \sin(\omega t)$, $\omega=2\pi f$, $T=1/f$

### AC Power and Energy

Power factor=$\cos\varphi$

$P=VI\cos\varphi$, $Q=VI\sin\varphi$, $S=VI$

3-phase $P=\sqrt{3}\, V_L I_L \cos\varphi$

Energy=Power×Time, 1 unit=1 kWh

### Capacitor and Transformer

Parallel C: $C_{eq}=C_1+C_2+...$

Series C: $1/C_{eq}=1/C_1+1/C_2+...$

$N_2/N_1=V_2/V_1$; $V_p/V_s=N_p/N_s$

ELECTRICAL UNITS AND SYMBOLS BANK

## Symbols and units

| Quantity | Symbol | Unit |
|---|---|---|
| Current | I | A |
| Voltage | V | V |
| Resistance | R | Ω |

| | | |
|---|---|---|
| Power | P | W |
| Energy | E | J/kWh |
| Charge | Q | C |
| Frequency | f | Hz |
| Time period | T | s |
| Angular velocity | ω | rad/s |
| Amplitude | Vm | V |
| Reactance | X | Ω |
| Impedance | Z | Ω |
| Power factor | cosφ | No unit |
| Active power | P | W |
| Reactive power | Q | VAR |
| Apparent power | S | VA |
| Capacitance | C | F |
| Inductance | L | H |
| Turns ratio | N2/N1 | No unit |

**CIRCUIT DIAGRAM BANK**

# Circuits and waveforms to practise

## Module 1

- Simple DC circuit
- Ohm’s law circuit
- Series/parallel/series-parallel resistor circuits
- AC waveform

### Module 2

- Residential wiring block
- Power triangle
- Energy meter block

### Module 3

- Capacitor charging/discharging
- Capacitor combinations
- Transformer diagram

### Module 4

- Diode and zener symbols
- Rectifier circuits and waveforms
- NPN/PNP symbols
- Logic gate symbols

COMPONENT IDENTIFICATION BANK

## Identify components

### Resistors

Fixed, variable, carbon composition, metal film, wire wound; identified by bands and rating.

### Capacitors

Ceramic, polyester, electrolytic, variable; electrolytic has polarity marking.

### Inductors

Air core, iron core, ferrite core, variable; used in filters and chokes.

### Transformer

Primary, secondary, core, voltage/current rating.

### Diode / zener

Cathode band and part markings.

### Transistor and IC

Packages such as TO-92; logic ICs have VCC and GND pins.

SOLVED NUMERICAL EXAMPLES

## Complete calculation practice

**Power calculation**

V=12 V, R=4 Ω. I=V/R=12/4=3 A. P=VI=12×3=36 W.

**Series-parallel resistance**

R1=4 Ω and R2=6 Ω in series: 10 Ω. This is parallel with R3=5 Ω: 1/Req=1/10+1/5=3/10, Req=3.33 Ω.

**AC terms**

T=0.02 s. f=1/T=50 Hz. $\omega=2\pi f=314$ rad/s.

**Single phase power**

V=230 V, I=8 A, pf=0.75. $P=VI\cos\varphi=230\times8\times0.75=1380$ W=1.38 kW.

**Monthly bill**

60 W lamp used 5 h/day for 30 days at ₹5/unit. Power=0.06 kW, time=150 h, energy=9 units, charge=₹45.

**Transformer turns ratio**

$N_p=500$, $N_s=100$, $V_p=230$ V. $V_s=V_p\times N_s/N_p=230\times100/500=46$ V.

EXPECTED QUESTIONS

# Module-wise question bank

## Module 1

**Very short**

1. Define current, voltage and resistance.
2. State Ohm’s law.
3. Define frequency and time period.

**Numerical**

1. Use V=IR.
2. Find equivalent resistance.
3. Find f, T and ω.

# Module 2

**Very short**

1. What is power factor?
2. Define active power.
3. What is ELCB?

**Numerical**

1. Calculate AC power.
2. Calculate electricity bill.
3. Calculate three phase power.

# Module 3

**Very short**

1. Define resistor and capacitor.
2. What is mutual inductance?
3. What is turns ratio?

**Numerical**

1. Resistor colour code.
2. Capacitor combinations.
3. Transformer secondary voltage.

# Module 4

**Very short**

1. What is rectification?
2. What is zener diode?
3. Name universal gates.

**Circuit / truth table**

1. Draw rectifier circuits.
2. Write all logic gate truth tables.

## Objective / MCQ

| No. | Question | Options |
|---|---|---|
| 1 | Ohm’s law is: | A P=VI B V=IR C R=ρl/A D v=Vm sinωt |
| 2 | Unit of resistance: | A volt B ampere C ohm D watt |
| 3 | Commercial unit of energy: | A watt B joule C kWh D ampere |
| 4 | Transformer voltage changes by: | A resistance B turns ratio C colour code D logic level |
| 5 | NAND and NOR are: | A rectifiers B amplifiers C universal gates D passive gates |

PRACTICE PROBLEMS

# Module-wise practice with final answers

### Module 1

1. V=48 V, R=12 Ω. Find I. **Ans: 4 A**
2. 2Ω, 4Ω, 6Ω series. **Ans: 12 Ω**
3. Time period of 50 Hz AC. **Ans: 0.02 s**

### Module 2

1. V=230 V, I=4 A, pf=0.8. **Ans: 736 W**
2. 2 kW load, 3 h/day, 30 days. **Ans: 180 units**
3. 3-phase: VL=400 V, IL=5 A, pf=0.8. **Ans: 2.77 kW**

### Module 3

1. Red, Violet, Brown, Gold. **Ans: 270 Ω ±5%**
2. 10 μF and 10 μF series. **Ans: 5 μF**
3. Vp=240, Np=1000, Ns=200. **Ans: 48 V**

## Module 4

1. Draw bridge rectifier and waveform.
2. Write XOR truth table.
3. Explain transistor as amplifier.

## Self-assessment

- ✓ I can solve Ohm’s law.
- ✓ I can calculate kWh and bill.
- ✓ I can draw rectifiers.
- ✓ I can complete truth tables.

QUICK REVISION

# Last-day revision

## Module 1

- $V=IR$
- $P=VI=I^2R=V^2/R$
- Series adds
- Parallel reciprocal adds
- $v=V_m \sin\omega t$

## Module 2

- $PF=\cos\varphi$
- $P=VI\cos\varphi$
- $Q=VI\sin\varphi$
- $S=VI$
- 1 unit=1 kWh

## Module 3

- Colour code
- Capacitors add in parallel
- Reciprocal in series
- Transformer: mutual induction

## Module 4

- Diode conducts forward
- Rectifier AC to DC
- Zener regulates
- Transistor amplifies
- NAND/NOR universal

**Common Mistakes:** not converting W to kW in energy problems, adding parallel resistors directly, mixing capacitor and resistor formulas, ignoring tolerance band, and writing incomplete truth tables.

MODEL QUESTION PAPER WITH FULL ANSWERS

# FUNDAMENTALS OF ELECTRICAL AND ELECTRONICS ENGINEERING • 75 Marks • 3 Hours

## PART A - Answer all. [10 × 1 = 10]

1. State Ohm’s law.
2. Write the unit of resistance.
3. Write the equation of AC voltage.
4. Define power factor.
5. What is commercial unit of energy?
6. Name the device used to measure house energy consumption.
7. What is an electrolytic capacitor?
8. State transformer working principle.
9. What is rectification?
10. Name two universal gates.

## PART B - Answer any five. [5 × 5 = 25]

1. A 12 Ω resistor is connected across 24 V. Find current and power.
2. Derive equivalent resistance of series resistors.
3. Compare series and parallel circuits.
4. A single phase load takes 6 A from 230 V at 0.8 pf. Find active and apparent power.
5. A 1.5 kW heater works 2 h daily for 30 days. Calculate units consumed.
6. Find value of resistor bands Red, Violet, Brown, Gold.
7. Write NAND truth table and why it is universal.

## PART C - Answer any four. [4 × 10 = 40]

1. Explain AC generation and define cycle, frequency, time period, angular velocity, amplitude, reactance and impedance.
2. Explain residential wiring components and safety precautions.
3. Draw power triangle and calculate P,Q,S for V=230 V, I=10 A, pf=0.6.
4. Explain capacitors and combinations. Find equivalent capacitance of 4 μF and 12 μF in series and parallel.
5. Explain transformer working. Np=1000, Ns=200, Vp=250 V. Find Vs.
6. Explain half wave and bridge rectifiers with waveform.
7. Write symbols and truth tables of AND, OR, NOT, NAND, NOR and XOR gates.

ANSWER KEY / COMPLETE SOLUTIONS

# Full model paper answers

## Part A

| No. | Answer |
|---|---|
| 1 | At constant temperature, current is directly proportional to voltage: V=IR. |
| 2 | Ohm (Ω). |
| 3 | v=Vm sin(ωt). |
| 4 | Power factor is cosφ. |
| 5 | kilowatt-hour (kWh). |
| 6 | Energy meter. |

| | |
|---|---|
| 7 | Polarized high-capacitance capacitor with polarity marking. |
| 8 | Mutual electromagnetic induction. |
| 9 | Conversion of AC into pulsating DC. |
| 10 | NAND and NOR. |

## Part B Solutions

**B1**

R=12 Ω, V=24 V. I=V/R=24/12=2 A. P=VI=24×2=48 W.

**B2**

In series same current flows. V=V1+V2+V3. Using Ohm law: IReq=IR1+IR2+IR3. Hence Req=R1+R2+R3.

**B3**

Series: same current, voltage divides, resistance adds, one open stops circuit. Parallel: same voltage, current divides, equivalent resistance less than smallest, domestic loads use parallel.

**B4**

V=230 V, I=6 A, pf=0.8. S=VI=1380 VA. P=VIcosφ=1104 W.

**B5**

Power=1.5 kW. Time=2×30=60 h. Energy=1.5×60=90 kWh=90 units.

**B6**

Red=2, Violet=7, Brown=×10, Gold=±5%. Value=27×10=270 Ω ±5%.

**B7**

NAND truth table: 00→1, 01→1, 10→1, 11→0. It is universal because NOT, AND and OR can be built using only NAND.

## Part C Solutions

**C1 AC generation**

When a coil rotates in a magnetic field, flux linked with it changes and emf is induced. The voltage varies sinusoidally: $v=V_m \sin\omega t$. Cycle: one complete variation. Frequency: cycles/s. Time period: time for one cycle. Angular velocity: $\omega=2\pi f$. Amplitude: maximum value. Reactance: AC opposition due to L/C. Impedance: total AC opposition. Malayalam: AC voltage positive-negative □□□ □□□□□□□□□□□□□ sine waveform □□□□□□□□□.

**C2 Wiring and safety**

Service connection supplies consumer installation. Cut-out fuse protects overcurrent. Energy meter measures kWh. Main switch controls supply. MCB protects overload/short circuit. ELCB protects leakage shock. Neutral link connects neutrals. Safety: switch off supply, use correct ratings, proper earthing, avoid wet hands, do not overload sockets, use insulated tools. Malayalam: MCB equipment protect □□□□□□□□, ELCB shock hazard □□□□□□□□□□□.

**C3 Power triangle**

$P=VI\cos\varphi$, $Q=VI\sin\varphi$, $S=VI$. Given $V=230$, $I=10$, $\cos\varphi=0.6$ so $\sin\varphi=0.8$. $S=2300$ VA. $P=1380$ W. $Q=1840$ VAR.

**C4 Capacitors**

Capacitor stores charge in electric field. Charging voltage rises; discharging voltage falls. Uses: filtering, coupling, timing, smoothing. For 4 μF and 12 μF: Parallel=16 μF. Series: $1/C=1/4+1/12=1/3$, so $C=3$ μF.

### C5 Transformer

Transformer works by mutual induction between primary and secondary windings through magnetic core. Vs/Vp=Ns/Np. Vs=250×200/1000=50 V. Step-down transformer. Malayalam: Secondary turns □□□□□□□□□□□□□ voltage □□□□□□□.

### C6 Rectifiers

Half wave rectifier uses one diode. It conducts during one half cycle only. Bridge rectifier uses four diodes; two conduct each half cycle, so load current remains same direction for both halves. Bridge gives better output and needs no centre tap.

### C7 Logic gates

AND: 00=0,01=0,10=0,11=1. OR: 00=0,01=1,10=1,11=1. NOT: 0=1,1=0. NAND: 00=1,01=1,10=1,11=0. NOR: 00=1,01=0,10=0,11=0. XOR: 00=0,01=1,10=1,11=0. Applications: computers, calculators, alarms, control and automation.

## MCQ Answer Key

1. B - V=IR. 2. C - ohm. 3. C - kWh. 4. B - turns ratio. 5. C - universal gates.

**REFERENCES AND ONLINE RESOURCES**

# Official learning resources

| Code | Book Title / Author |
|---|---|
| T1 | B. L. Theraja and A. K. Theraja, Textbook of Electrical Technology: Part 1 - Basic Electrical Engineering in S. I. Units, S. Chand Publication, 2012. |
| T2 | N. N. Bhargava, D. C. Kulshreshtha, S. C. Gupta, Basic Electronics and Linear Circuits, Tata McGraw Hill Education, 1983. |
| R1 | A. Anand Kumar, Fundamentals of Digital Circuits, Prentice Hall India Pvt. Ltd., 2014. |

## Formula checklist

- ✓ I can use V=IR and power formulas.
- ✓ I know resistance formulas.
- ✓ I can calculate AC power and bill.
- ✓ I can solve capacitor combinations.
- ✓ I can use transformer turns ratio.

## Diagram checklist

- ✓ I can draw simple, series and parallel circuits.
- ✓ I can draw wiring block.
- ✓ I can draw rectifiers and waveforms.
- ✓ I can draw transistor symbols and logic gates.
- ✓ I can write truth tables.

**Final Tip:** A complete answer usually needs formula, circuit/diagram or waveform, explanation and final units.